

AskIt Research Lab — Home Assignment Submission

Repository (code, data, full logs, commit history): <https://github.com/DeanData/askit-persona-function>

Metric, committed before any text was generated:

<https://github.com/DeanData/askit-persona-function/blob/main/METRIC.md>

The metric was committed in METRIC.md before any text was generated. This report summarizes the work; full detail is in the repository.

The function

`f(audience description, persona description, context) -> text`. Audiences: 5 finance- forum populations (r/personalfinance, r/wallstreetbets, r/fatFIRE, r/thetagang, r/povertyfinance), chosen over demographic-bin alternatives because they give a genuine two-attribute intersection (risk-orientation × sophistication) without inventing one — fatFIRE is the dev intersectional audience, thetagang (held out) mirrors its compound pattern but swaps the risk attribute. povertyfinance is held out as a single-attribute, demographically distinct population. Personas: a structured schema (age, age-conditioned occupation, region, family status, one open-ended detail), rendered through one fixed template, 40 per audience, fixed seed, shared roster across every condition (paired comparison). No persona field restates the audience's defining trait. Contexts: one structural ("\$10k advice-giving," forces justification/argument) and one surface ("weekly money check-in," low-stakes, lets register surface unforced) — deliberately chosen to pull on the function differently, both finance-anchored (checked against the generality requirement; audiences, not topics, are what must generalize).

Metric (committed in METRIC.md before any text was generated)

Ten cheap stylometric features (6 structural: sentence length, lexical diversity, word length, first-person rate, hedge rate, punctuation density; 4 surface: emoji, exclaim, all-caps, meme-token rate) — validated empirically in a Phase 1 pre-check on real data (5-class nearest-centroid accuracy 42.0% full vs. 41.6% structural-only vs. 20% chance: separation is structural, not slang-driven). Ground truth is real Reddit text, split into a disjoint eval pool (242/audience, used only for reference statistics) and a few-shot pool (8/audience, used only in prompts) — a contamination risk caught before it was built, not after. Success is defined as **matching real audiences, not maximizing separation**: an earlier draft treated a bigger between/within ratio as better, which directly rewards within-audience collapse; both fidelity measures are now signed relative error against real references (0 = perfect match). Self-labeled texts (literal audience name-dropping) are excluded from fidelity, counted separately. Bootstrap 95% CIs (n=1000) on every number. **What a perfect score would still miss**: style, not content; personalfinance/povertyfinance are known-hard to separate on this feature set; the meme-token wordlist is crude; personas are demographically uniform by design, so some fidelity gap may reflect demographic mismatch rather than voice failure.

Numbers (see artifacts/baseline_vs_fewshot.png)

Baseline (naive prompt: audience + persona + context, nothing else) shows a consistent homogenization pattern on dev: within-variation fidelity negative for all 3 dev audiences (personalfinance −0.17, wallstreetbets −0.43, fatFIRE −0.12; all CIs exclude zero), centroid drift substantial (0.37–0.52), between-separation also negative (−0.20). The baseline is more homogeneous than real text on every axis at once.

Fewshot (8 real per-audience examples added as a style primer) was evaluated against this on the same paired roster. On dev, the result is **mixed-to-negative**: within- variation flat in aggregate (MARE 0.242→0.243) and worse for fatFIRE specifically (−0.12→−0.15); between-separation moved away from zero (−0.20→−0.25, worse); centroid drift improved only marginally, driven almost entirely by fatFIRE. On held-out (thetagang, povertyfinance — run

exactly once, after dev was finalized), the picture flips: every measure moves in the favorable direction (within-variation, centroid drift, and between-separation all improve). But baseline/fewshot bootstrap CIs overlap for most held-out measures — a consistent directional signal, not a statistically decisive win. **Dev and held-out disagreed about whether the improvement helped** — exactly the disagreement the held-out mechanism exists to surface.

What didn't work

Few-shot's dev self-labeling rate more than doubled (6/240→13/240 texts), concentrated almost entirely in fatFIRE (3→12 of 80). Inspected the fatFIRE few-shot pool directly: 2 of its 8 real examples literally self-label ("Fatfire in unconventional cities," "... achieve fatFIRE?"). Showing the model real text that names its own community taught it to imitate that. This is a real, well-evidenced negative finding, not just a weak result. **Not fixed now:** filtering self-labeled examples from the few-shot pool and rerunning is exactly the "further tuning past the first defensible improvement" we agreed to cut if time was short — named here as a next step, not executed.

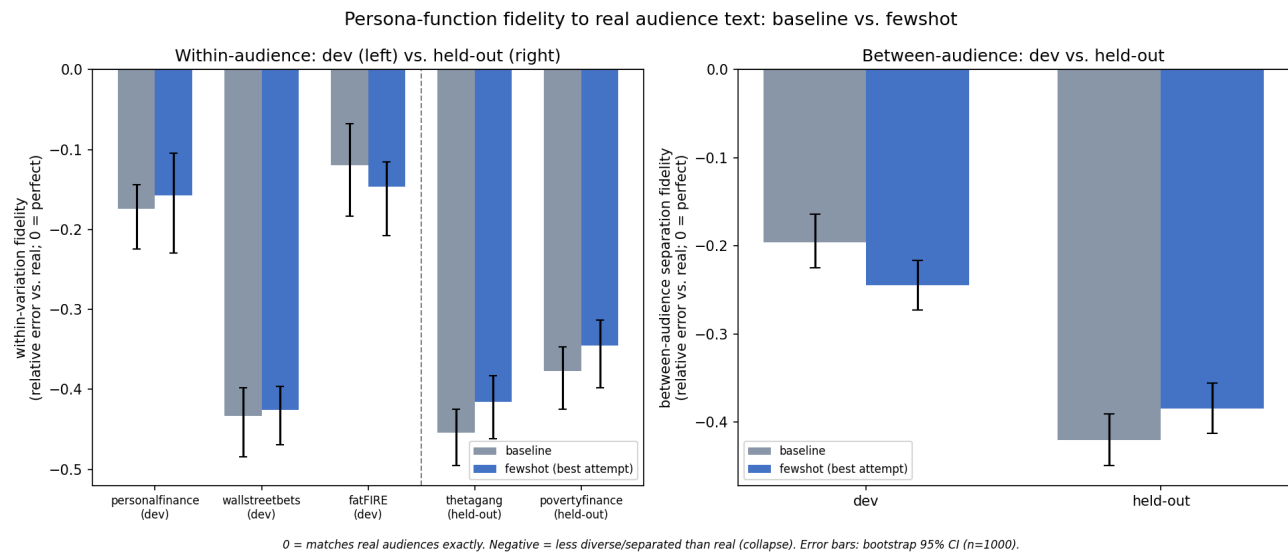
Part 3: Slow research

Placeholder — pending remaining time in the 5-hour budget. Working through the mechanism, cost, falsification test, and kill criterion collaboratively rather than proposing it cold; to be filled in if time allows, reported honestly as not reached otherwise.

What we'd do with another week

- Write a proper Part 3 proposal — a real non-prompt mechanism (steering vectors or a per-audience LoRA on a local model) — and then build and test it.
- Go after the within-audience collapse directly — it was the main failure — and test whether it's a limit of the prompt or of the model.
- Make the personas demographically realistic per audience, to separate demographic mismatch from real voice mismatch.
- Add a content/meaning axis to the metric, not just style, since matching how audiences write isn't the same as matching what they talk about.

Required visual



Where Claude Code was plausible but wrong

Early in Part 1, Claude Code proposed a 2×2 audience design (risk-orientation × sophistication) and labeled fatFIRE "the intersectional audience." It sounded coherent, but it was inconsistent: in a full 2×2, every audience is defined by both attributes, so none is uniquely the compound one — "intersectional" would have been true by construction, not by design, quietly erasing the single-vs-compound contrast the requirement asks for. The error surfaced when it was challenged directly — by that logic, aren't the other audiences equally intersections? The fix was to make risk-orientation the sole defining attribute for the single-attribute audiences, reserving the deliberate two-attribute combination for fatFIRE (dev) and thetagang (held-out). Caught and fixed early.

Audience and persona descriptions

The two inputs to `f()`. Submitted in the repository, not reproduced here: audience descriptions — [data/audiences.json](#); persona descriptions (200 personas, 40/audience) — [data/personas/](#).

Time log

Rough, net effort against the 5-hour budget (spread over more wall-clock than that, across multiple sessions).

- 1h — understanding the task and overall planning
- Working with Claude Code: 0.5h logistics/scaffolding, 1.5h Part 1, 1h Part 2, 1h deliverables

Net: ~5 hours.